

Borrower EMI Warning, Lock, and Unlock Lifecycle

Manager-facing explanation of when borrowers receive EMI warnings, when an EMI becomes overdue, when a phone enters grace period, when it locks, and how it can be unlocked.

Default Policy Item	Value
Grace period after missed due date	7 days
Lock after grace expiry	Enabled
EMI policy scan frequency	Every 30 minutes

If a tenant has custom policy settings, that tenant's configured grace period overrides this default.

Simple Timeline Example

Example: EMI due date is 10 July.

Date	What Happens
30 June	Borrower receives automatic reminder: EMI due in 10 days.
5 July	Borrower receives automatic reminder: EMI due in 5 days.
10 July	EMI is due.
11 July	EMI is treated as overdue if still unpaid or partially paid. Grace period starts.
11-17 July	Grace period is active; overdue reminders can be sent every 12 hours.
After 17 July	Phone becomes eligible for lock on the next EMI policy scan.

Because the EMI policy job runs every 30 minutes, warnings, grace updates, and locks happen on the next scheduled run after the threshold is crossed.

How Overdue Is Calculated

- An EMI is upcoming before its due date.
- An EMI becomes overdue when the due date has passed and the installment is still unpaid or partially paid.
- Outstanding amount is EMI amount plus penalty amount minus any amount already paid.

Borrower Warnings

Trigger	Borrower Warning
10 days before due date	Automatic upcoming EMI reminder.
5 days before due date	Automatic upcoming EMI reminder.
During grace period	Automatic overdue reminder every 12 hours, if the device has a valid notification token.
Manual tenant action	Tenant can manually send overdue EMI reminders from the tenant app.

Tenant App Visibility

- Upcoming EMI list shows borrowers with unpaid or partially paid EMIs due in the selected upcoming window.
- The default upcoming window is the next 10 days.
- The tenant can query a window from 1 to 365 days.
- The overdue list shows overdue installments, partially paid installments whose due date has passed, and pending installments whose due date has passed.
- Fully paid and waived installments do not appear as overdue.

When The Phone Enters Grace Period

The phone enters grace period when the EMI due date has passed and the borrower is still inside the tenant's grace window. With the default policy, grace starts after the due date is missed. During grace period, the phone is not fully locked; the system applies the grace policy and continues warning the borrower.

When The Phone Gets Locked

Lock Scenario	What Happens
EMI overdue beyond grace period	System automatically queues a lock.
Tenant manually locks device	Tenant admin queues a lock with a reason.
Temporary unlock expires	System relocks the device automatically.
Critical security or integrity risk	System can auto-lock if the tenant risk policy allows it.

EMI lock formula: Lock date = EMI due date + grace period days.

With default policy: Lock date = EMI due date + 7 days.

What Phone Locked Means

When locked, the device receives the EMI locked policy. Normal device use is restricted. The allowed apps are limited, including the EMI Shield app and dialer. Factory reset, status bar, and ADB are restricted.

How The Phone Gets Unlocked

Unlock Path	What Happens
Tenant approves borrower payment	Payment is applied to the EMI schedule and an unlock command is queued.
Tenant manually unlocks device	Tenant admin queues unlock with a reason.
Tenant approves borrower unlock request	Unlock command is queued after review.
Tenant grants temporary unlock	Device unlocks for a limited duration.
Partner resolves escalated case	Partner can unlock or temporarily unlock after escalation.

Important: the backend queues the unlock, but the phone state changes only after the borrower device receives and acknowledges the command.

Temporary Unlock

- Temporary unlock is used when limited access is needed without fully resolving the case.
- Default temporary unlock duration is 24 hours.
- Maximum temporary unlock duration is 72 hours by default.
- When temporary unlock expires, the system relocks the device automatically.

Borrower Unlock Request Flow

- A borrower can raise an unlock request when locked or facing a payment/device issue.
- The request first goes to the tenant.
- If the tenant does not resolve it within the configured SLA, it can escalate to partner.
- If the partner does not resolve it within their SLA, it can escalate to admin review.
- Tenant or partner can unlock, temporarily unlock, or reject the request.

Manager Summary

Stage	Expected Behavior
Before due date	Borrower gets 10-day and 5-day reminders.
After due date	EMI becomes overdue if unpaid or partially paid; grace period starts.
Grace period	Borrower receives overdue reminders every 12 hours.
After grace expires	Phone becomes eligible for lock.
Unlock	Payment approval, manual unlock, request approval, temporary unlock, or partner action can unlock the phone.